Declaration of AI assistance

Mathematics of Computation

Author-prepared disclosure; not an official publisher form or signature.

Manuscript. A certified sixteen-prime-factor lower bound for Lehmer's totient problem

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Claude through Claude Code assisted the original search programs, computational development, partial Lean work and drafting. OpenAI Codex assisted the ceiling and one-candidate reductions, targeted valuation and residue investigations, independent primality certificates, checking the stated finite reproductions, figures, editing and packaging. These efforts do not provide a full-conjecture proof. In this revision, Codex assisted the uniform factor-count construction, its proof, explicit-stack implementation, exact controls, title revision and submission files. The new appendix is supplied for author review and is not claimed as a full-conjecture proof.

Human oversight. The submitting author stated on 25 October 2026 that the earlier manuscript was a mix of human-written and AI-generated material with all parts checked and understood by a human. The added research material awaits author review. This records that author declaration. Human authors retain responsibility for mathematical accuracy, citations and the final version; AI systems are not human coauthors.

Verification scope. The repository contains partial Lean results for statements not depending on the exhaustive search. The full searches and the full Lehmer conjecture are not claimed to be Lean-formalised. The new uniform construction has a written proof and fresh exact controls; it is not Lean-formalized. The guarded k=14 run is explicitly incomplete.

Supporting record. The separate AI_PROVENANCE.md identifies the documented tasks and tools. No unprovided model version, human signature, confidential transcript or journal approval is asserted.

Policy reference. https://www.ams.org/publications

Prepared on 25 October 2026. The author reviews this statement and completes any additional live-portal declarations before submission.